



Lab 01

GitHub Identity, Student Portal Registration,
and Ubuntu Server Installation

Student Name: Laiba Rani

Registration No: 2023-BCS-053

Course: CS-649 — Cloud DevOps

Section: A

GitHub Username: laibarani899-cpu



Task 1: Create and Configure Your GitHub Account

The screenshot shows the GitHub profile of user 'laibarani899-cpu'. The profile includes a circular avatar with a purple and white pixelated design. The name 'Laiba Rani' is displayed below the avatar, along with the username 'laibarani899-cpu' and the pronouns 'she/her'. An 'Edit profile' button is visible. The email address 'laibarani899@gmail.com' is listed. The 'Popular repositories' section shows six public repositories: 'portfolio' (HTML), 'E-Commerce-Website' (HTML), 'CodeAlpha-task1' (CSS), 'CodeAlpha-task3' (CSS), 'CodeAlpha_Calculator' (JavaScript), and 'birthday' (HTML). The '26 contributions in the last year' section shows a calendar grid with green squares indicating contributions. The year '2026' is selected in the 'Contribution settings' dropdown.

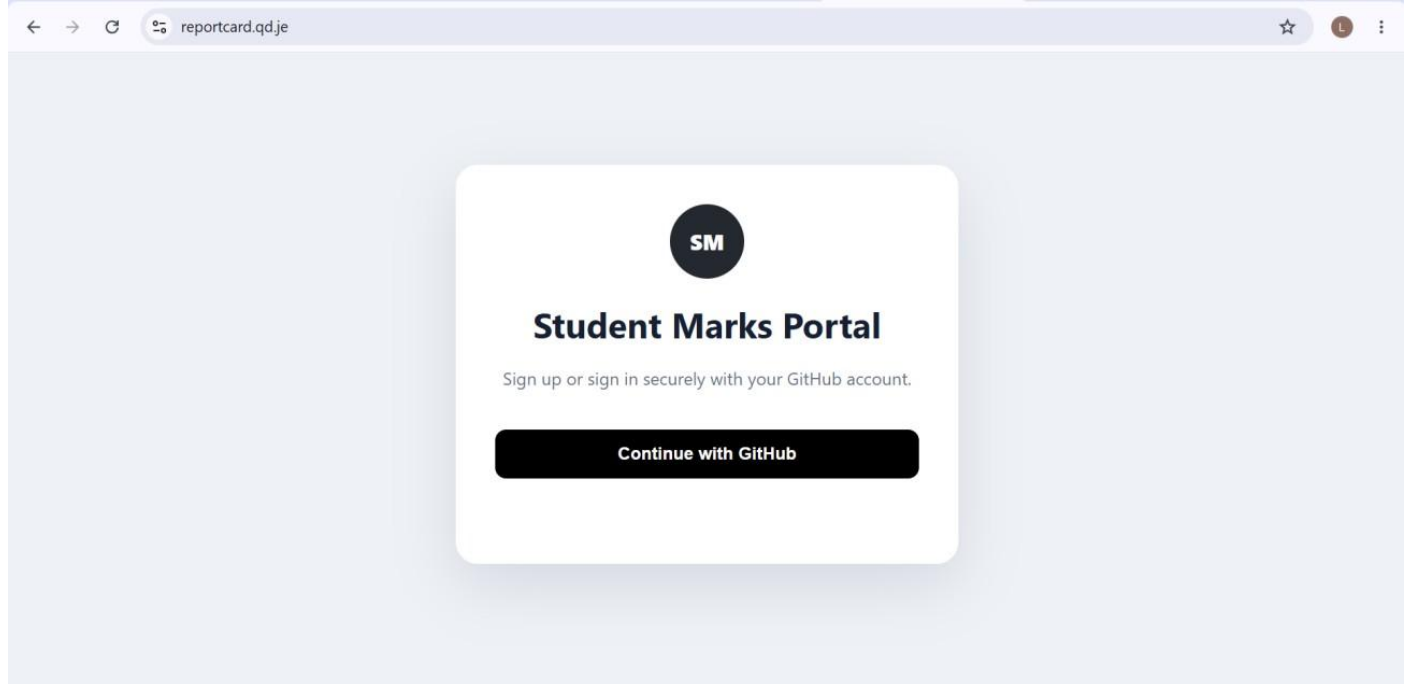
GitHub profile showing the configured professional username.

This screenshot is identical to the one above, showing the GitHub profile of user 'laibarani899-cpu'. It displays the profile information, including the name 'Laiba Rani', the username 'laibarani899-cpu', and the email 'laibarani899@gmail.com'. The 'Popular repositories' section lists six public repositories: 'portfolio' (HTML), 'E-Commerce-Website' (HTML), 'CodeAlpha-task1' (CSS), 'CodeAlpha-task3' (CSS), 'CodeAlpha_Calculator' (JavaScript), and 'birthday' (HTML). The '26 contributions in the last year' section shows a calendar grid with green squares indicating contributions. The year '2026' is selected in the 'Contribution settings' dropdown.

GitHub public profile showing the actual full name, Laiba Rani.



Task 2: Register in the Student Marks Portal



Student Marks Portal login page.



Authorize application - Google Chrome

github.com/login/oauth/authorize?response_type=code&clie...

Authorize Firebase GitHub Login

 **Firebase GitHub Login by Waqas Saleem**
wants to access your laibarani899-cpu account

 **Personal user data** 
Email addresses (read-only), profile information (read-only)

Cancel **Authorize WaqasSaleem97**

Authorizing will redirect to
<https://studentsreportcard-809ae.firebaseio.com>

 Not owned or operated by GitHub

 Created 3 months ago

 Fewer than 100 GitHub users

[Learn more about OAuth](#)



STUDENT MARKS PORTAL

Course enrollments

Sign out

**Laiba Rani**
2023-BSCS-53

+ Enroll in another course

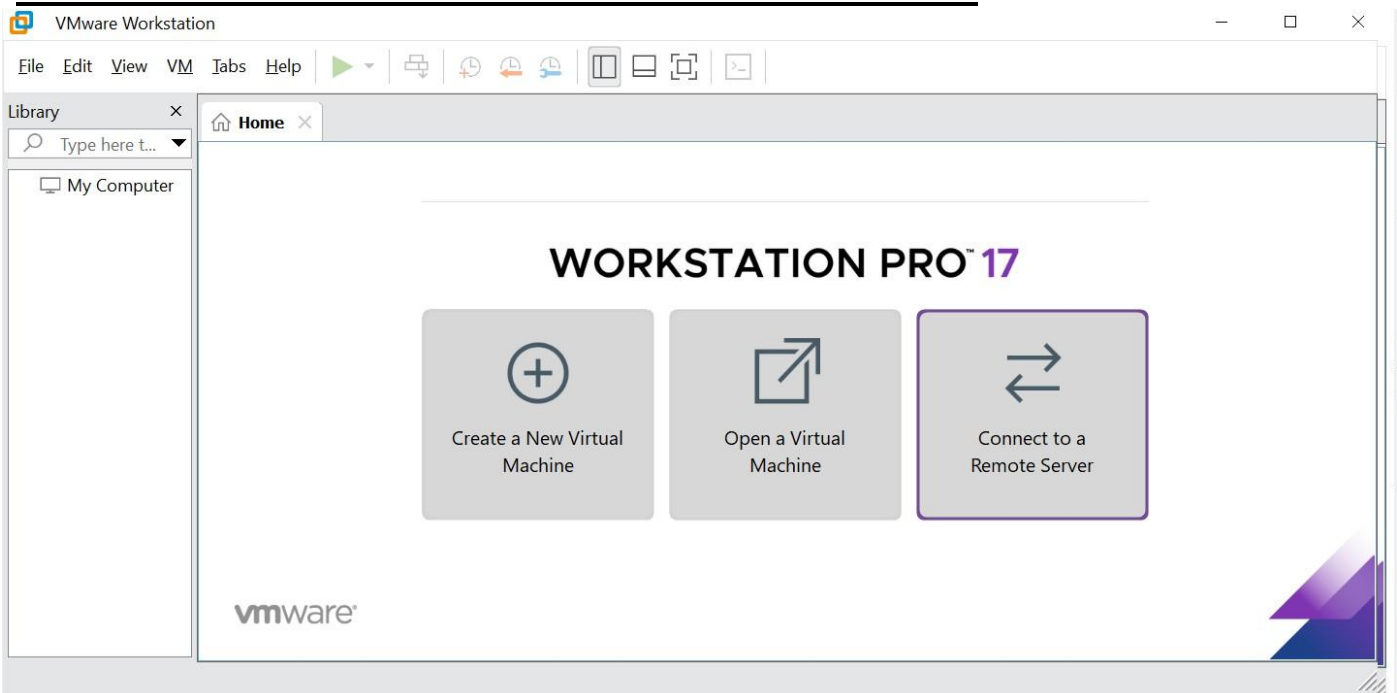
ENROLLMENT STATUS

Your courses

Cloud DevOps (CS-649) Section A
Waiting for administrator approval
Pending approval

Course enrollment submitted for CS-649 Cloud DevOps, Section A.

Task 3: Download VMware Workstation Pro and Ubuntu Server

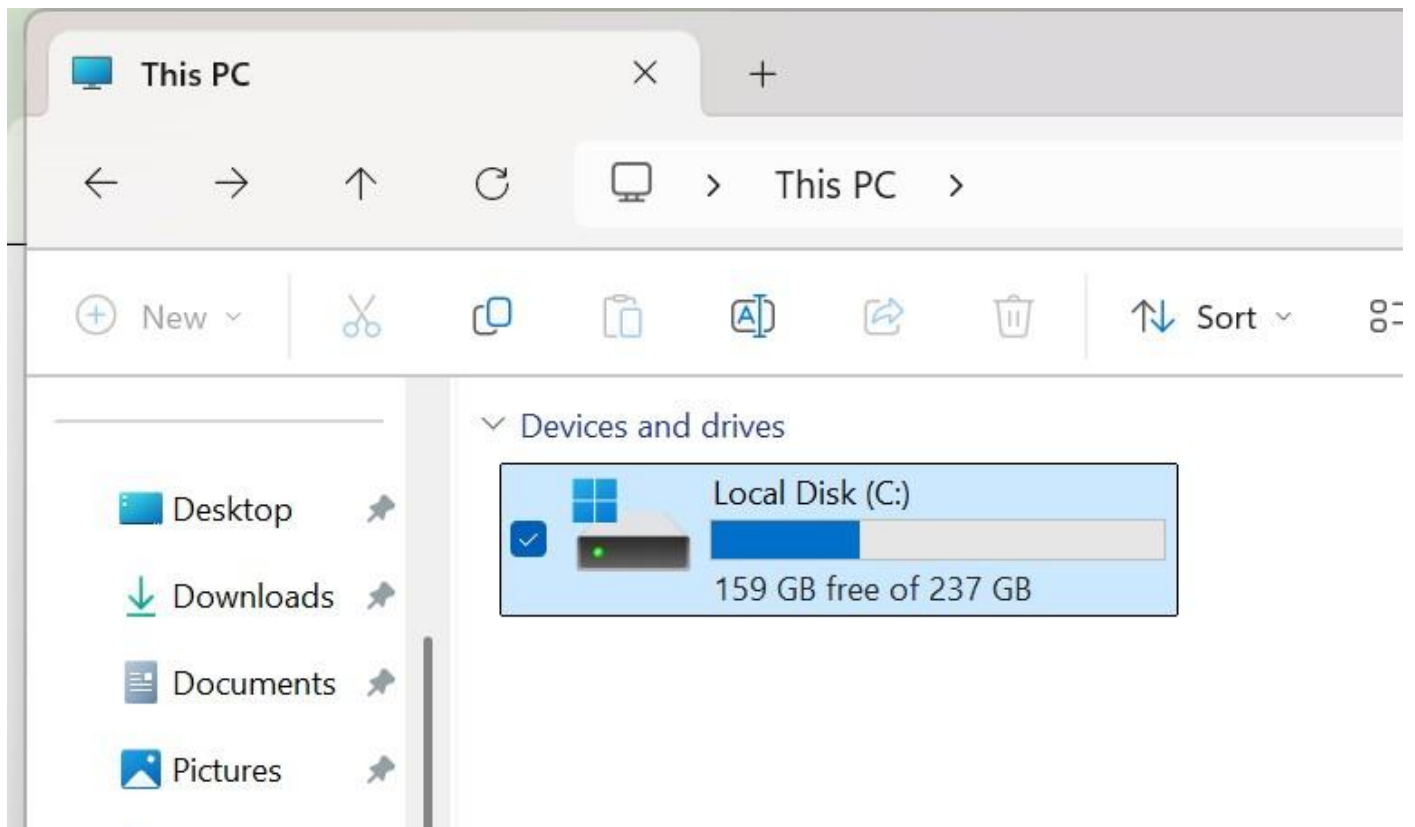


VMware Workstation Pro installed and opened.



☐ Name	Date modified	Type	Size
▼ Today			
☐ ubuntu-26.04.1-desktop-amd64	9/30/2026 11:07 AM	Disc Image File	6,330,478 KB
▼ Yesterday			
☐ VMware-workstation-full-17.6.4-24832109	9/29/2026 10:24 PM	Application	415,456 KB

Ubuntu ISO file downloaded on the computer.



Available storage on the Windows computer.

**Task 4: Create the Ubuntu Server Virtual Machine**

New Virtual Machine Wizard

Ready to Create Virtual Machine

Click Finish to create the virtual machine. Then you can install Ubuntu 64-bit.

The virtual machine will be created with the following settings:

Name:	Ubuntu 64-bit
Location:	C:\Users\PC\Documents\Virtual Machines\Ubuntu 64-bit
Version:	Workstation 17.5 or later
Operating System:	Ubuntu 64-bit
Hard Disk:	30 GB
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	2 CPU cores, CD/DVD, USB Controller, Sound Card

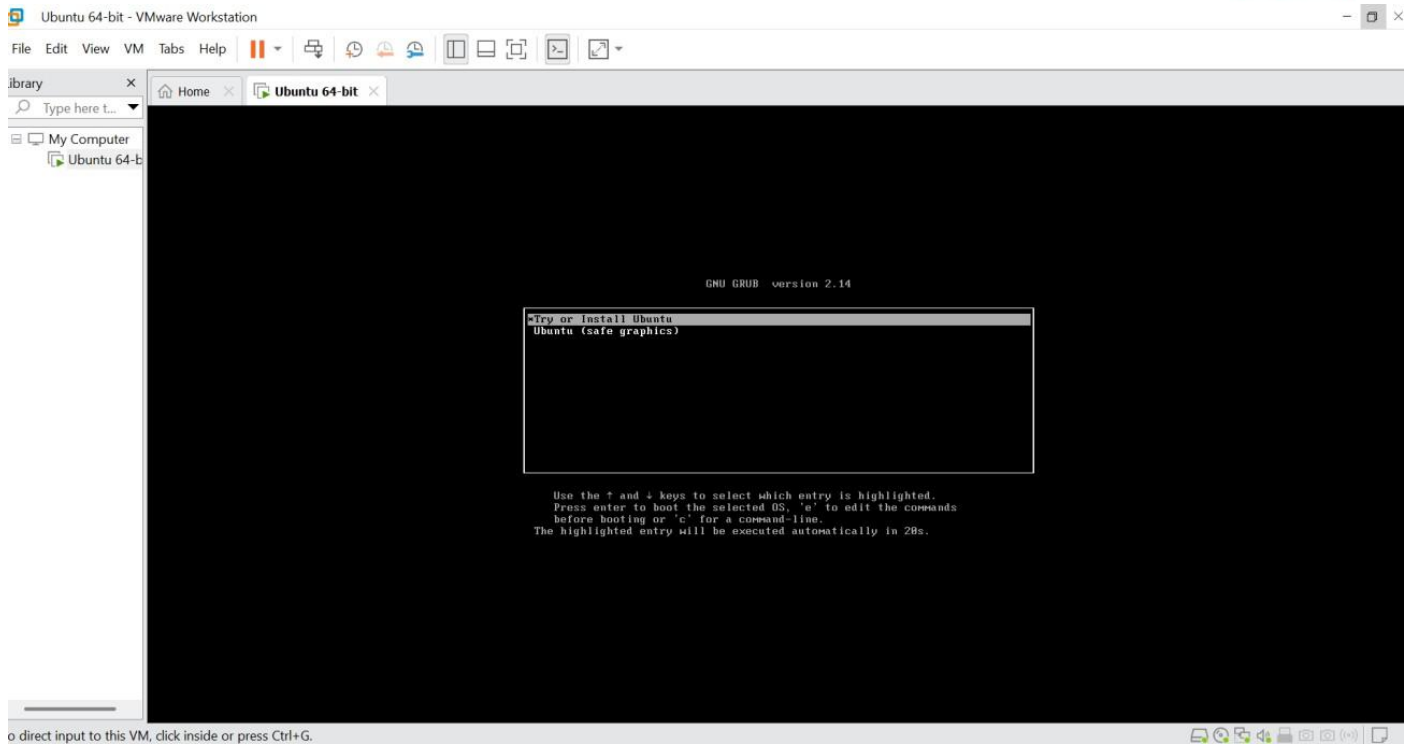
Customize Hardware...

< Back

Finish

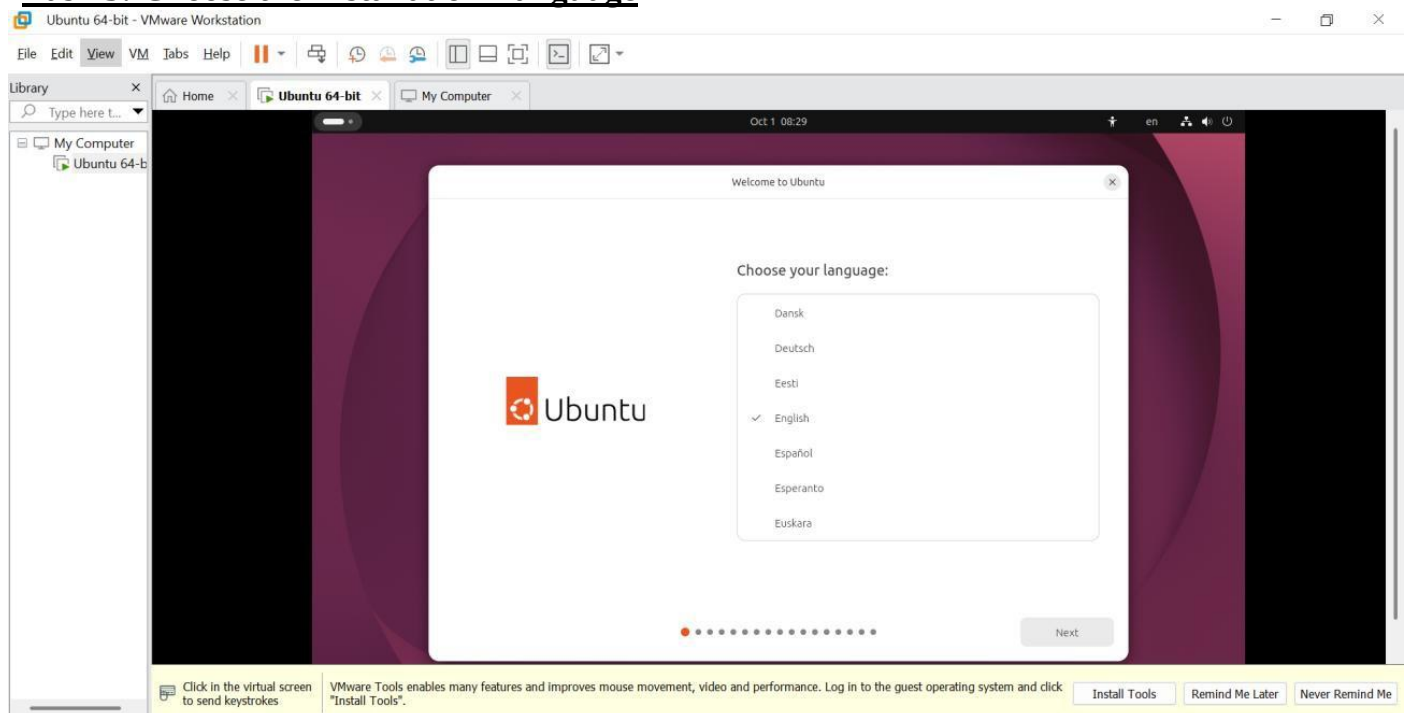
Cancel

VMware virtual machine configuration summary before creation.



Ubuntu installer boot screen.

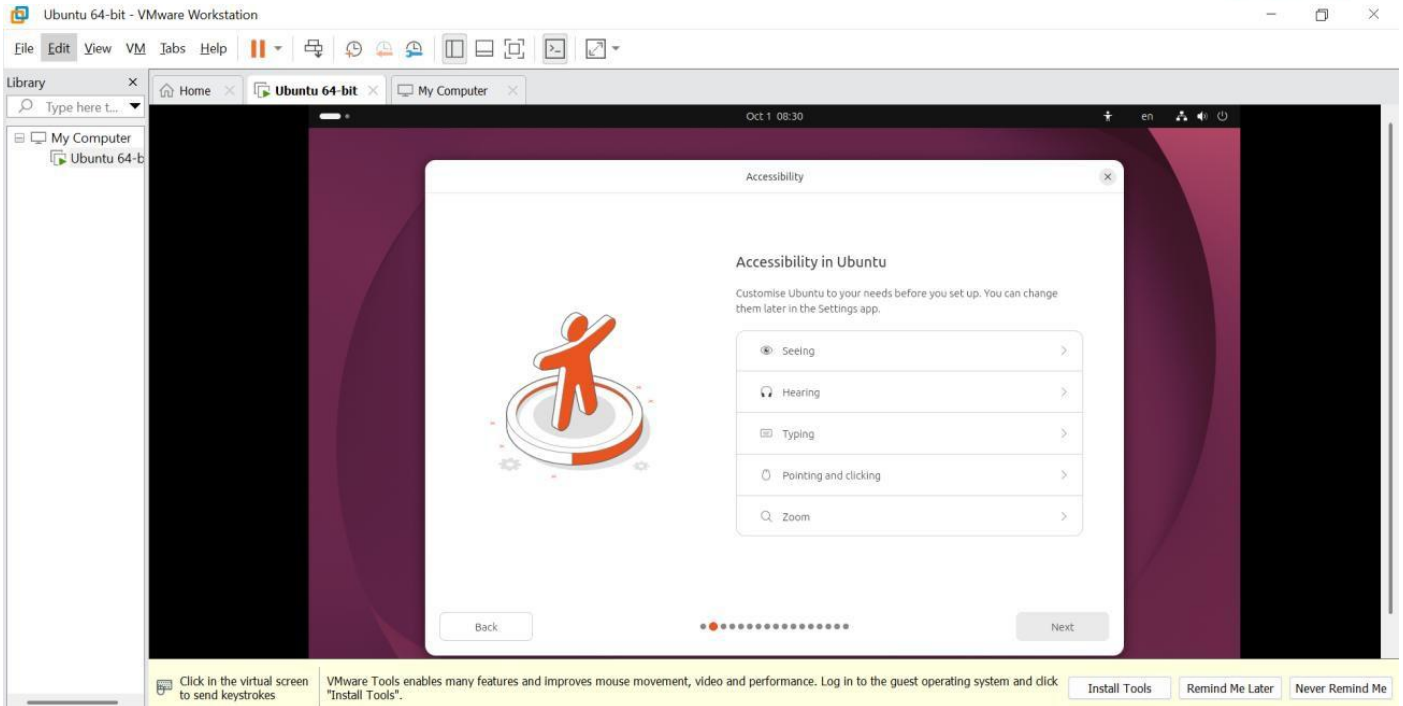
Task 5: Choose the Installation Language



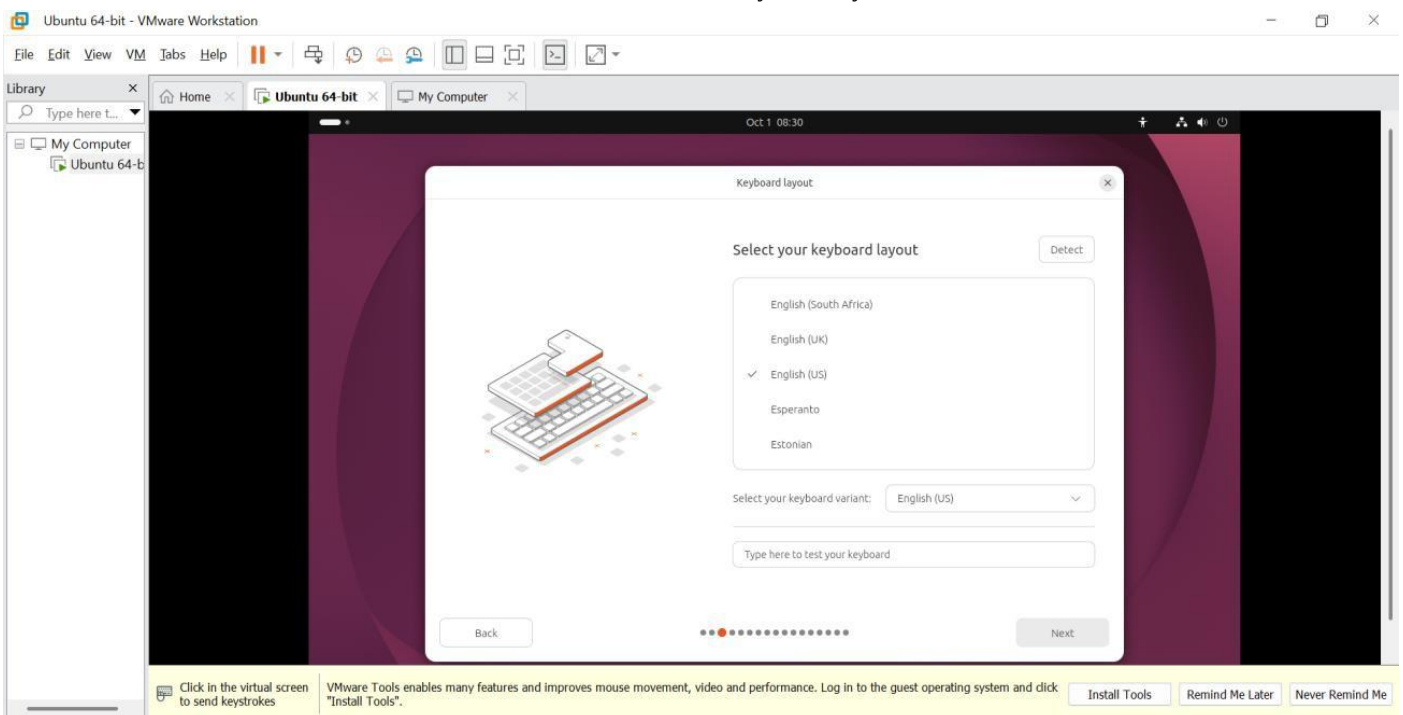
Ubuntu installer language selection screen.



Task 6: Configure the Keyboard Layout



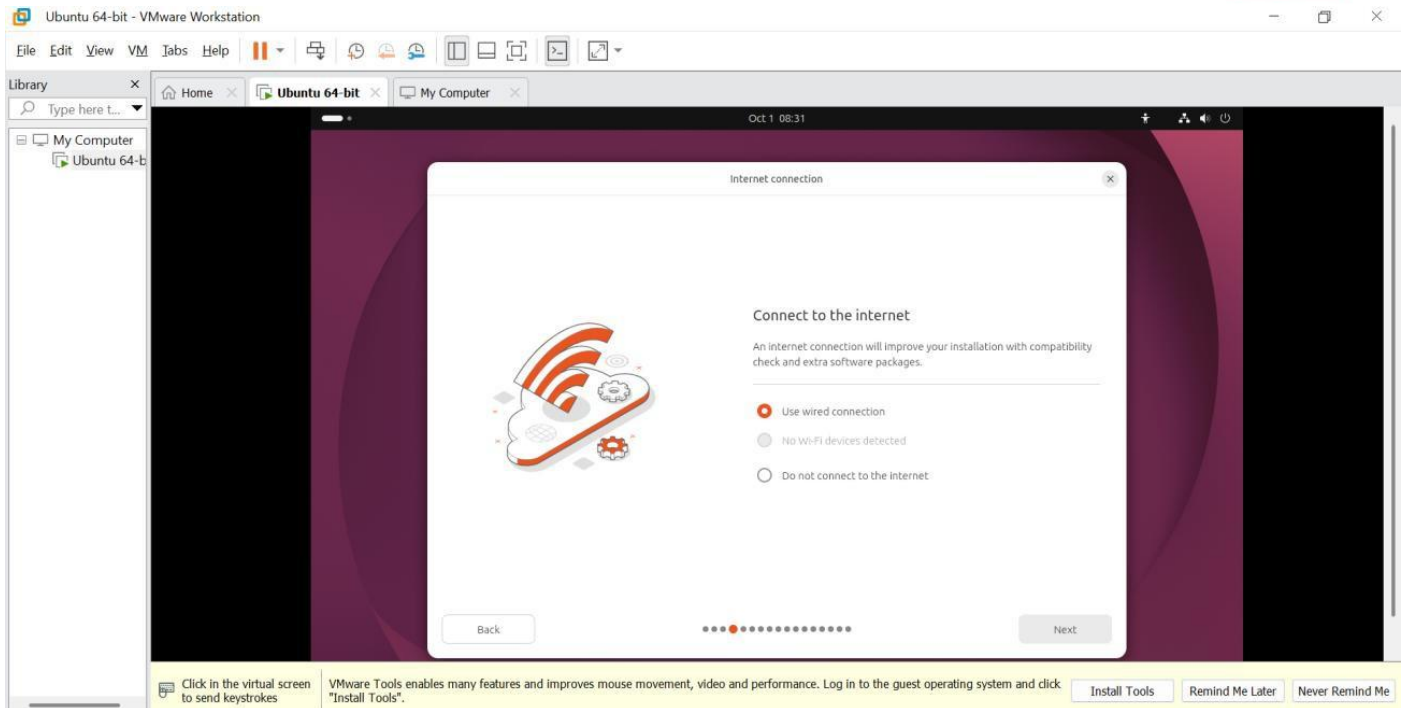
Ubuntu installer keyboard layout selection.



Ubuntu installer keyboard variant selection.

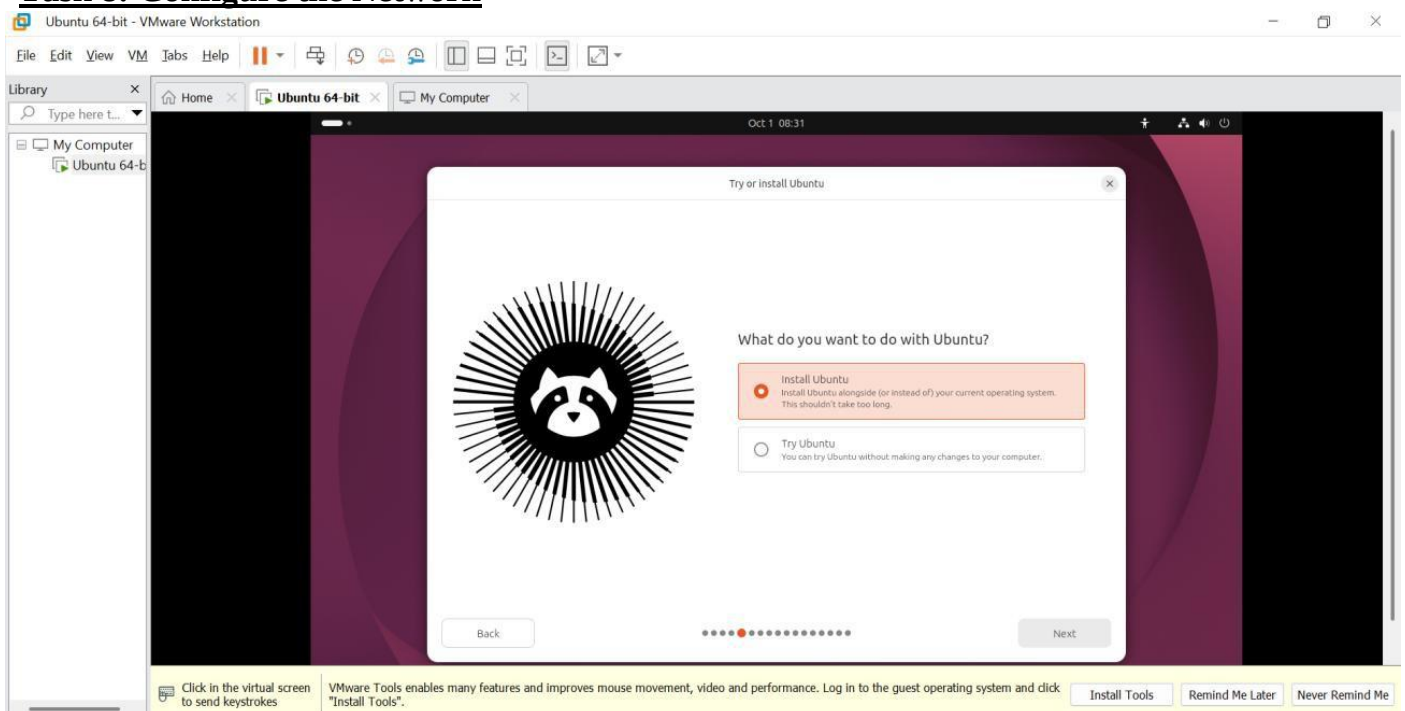


Task 7: Choose the Ubuntu Server Installation Type

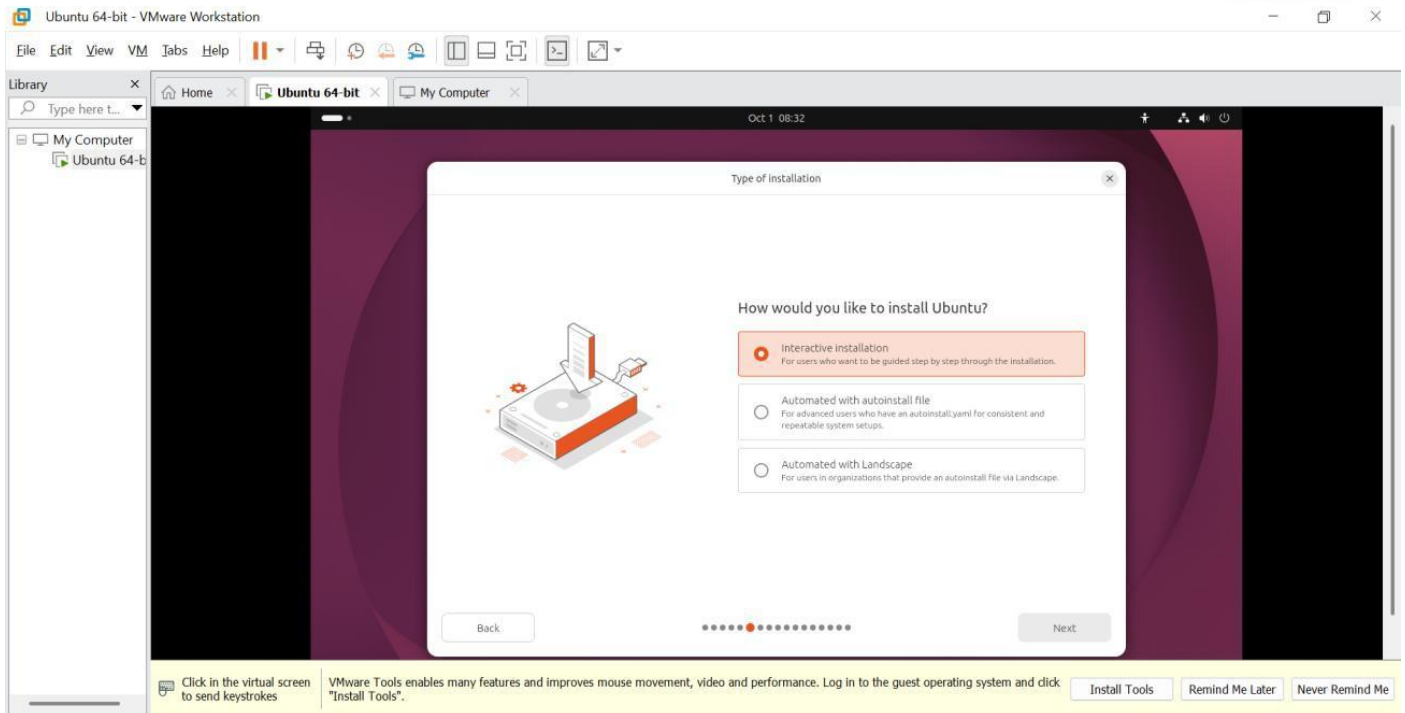


Ubuntu installation type screen.

Task 8: Configure the Network

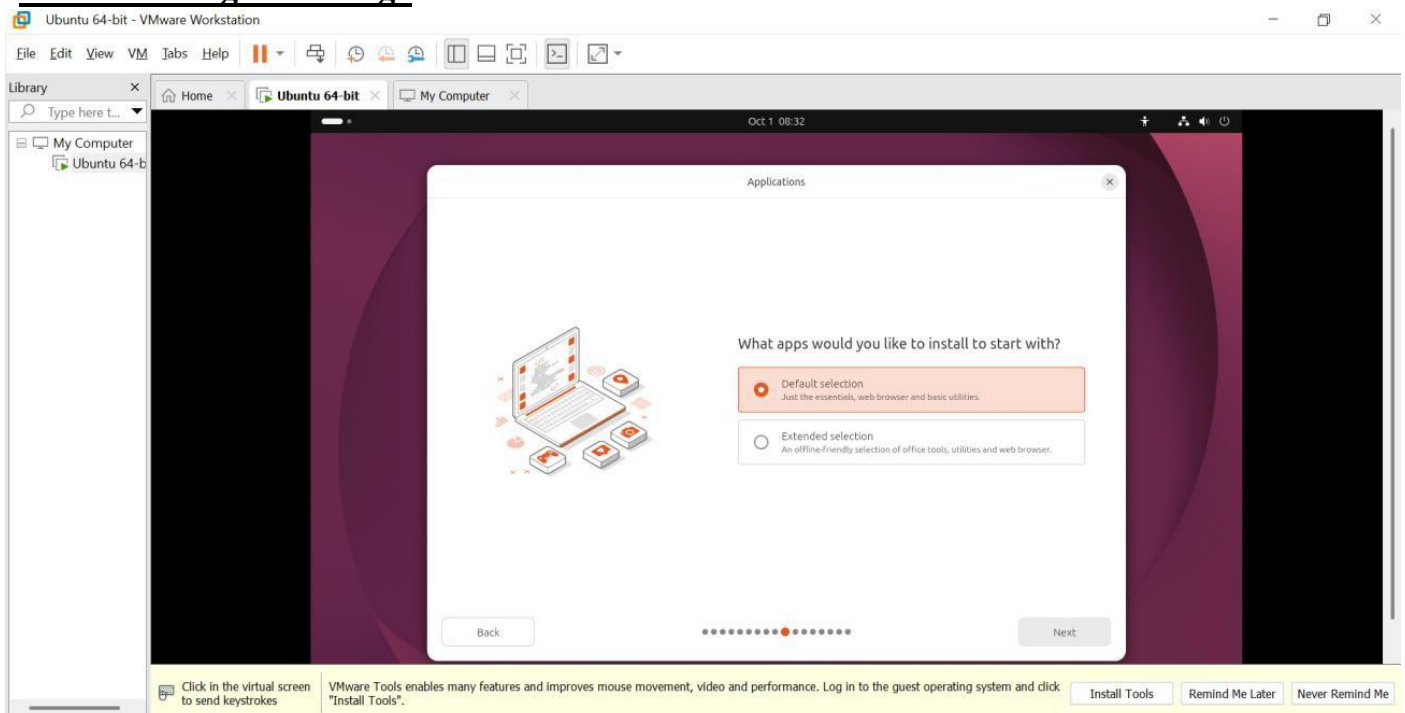


Ubuntu installer network interface screen.

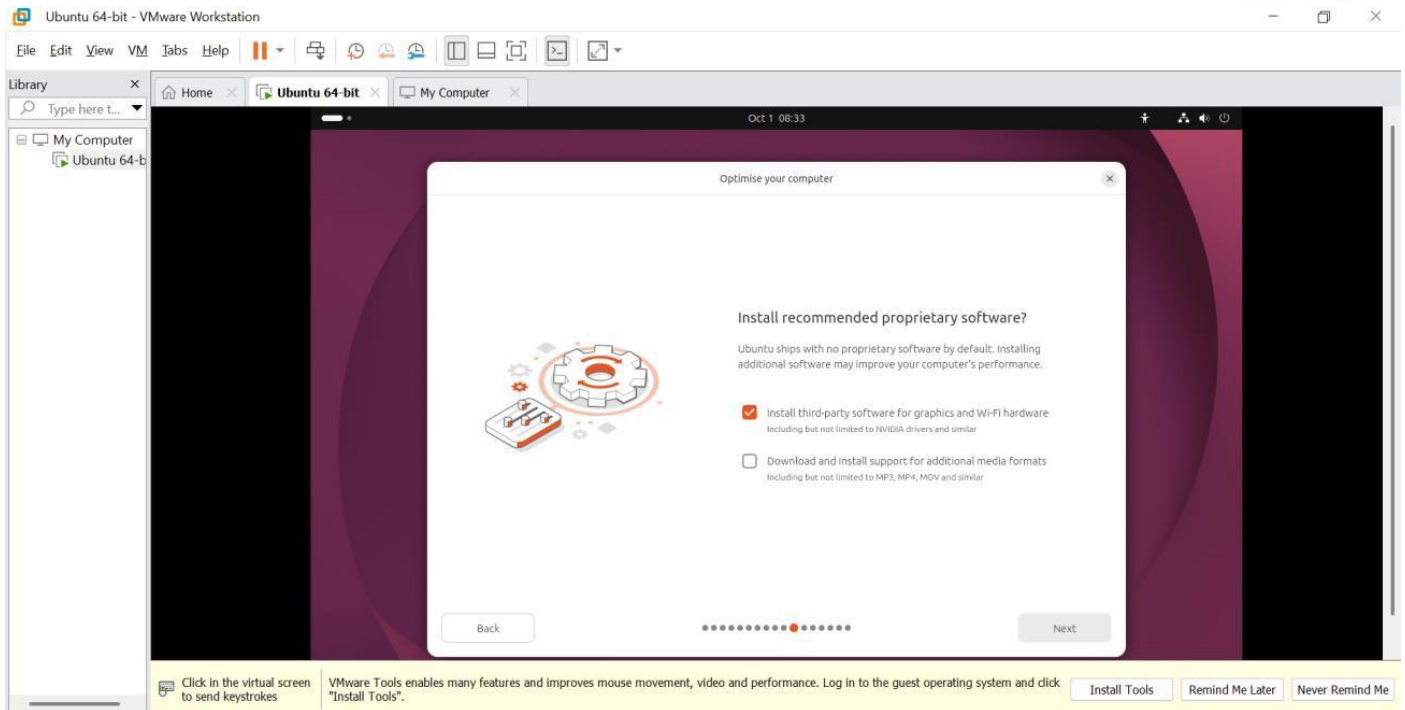


Ubuntu installer network/IP information.

Task 9: Configure Storage

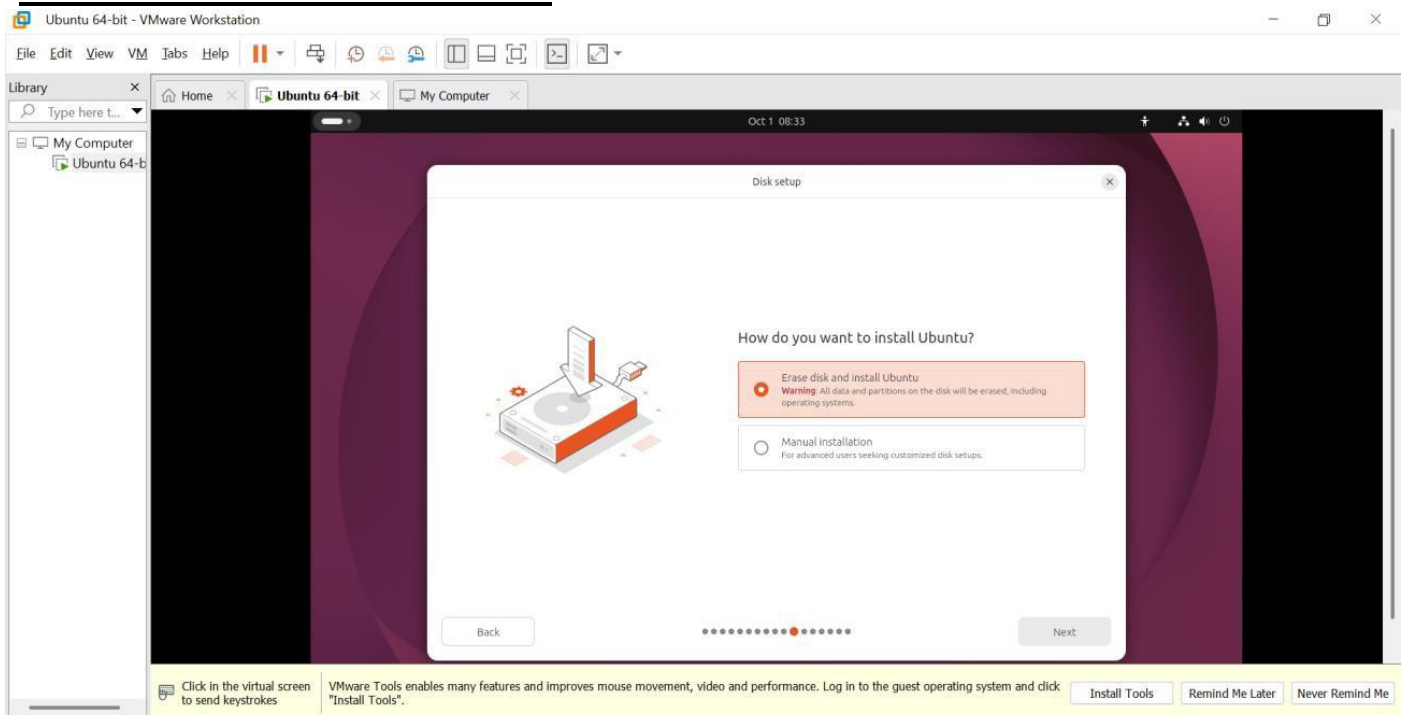


Ubuntu guided storage configuration screen.

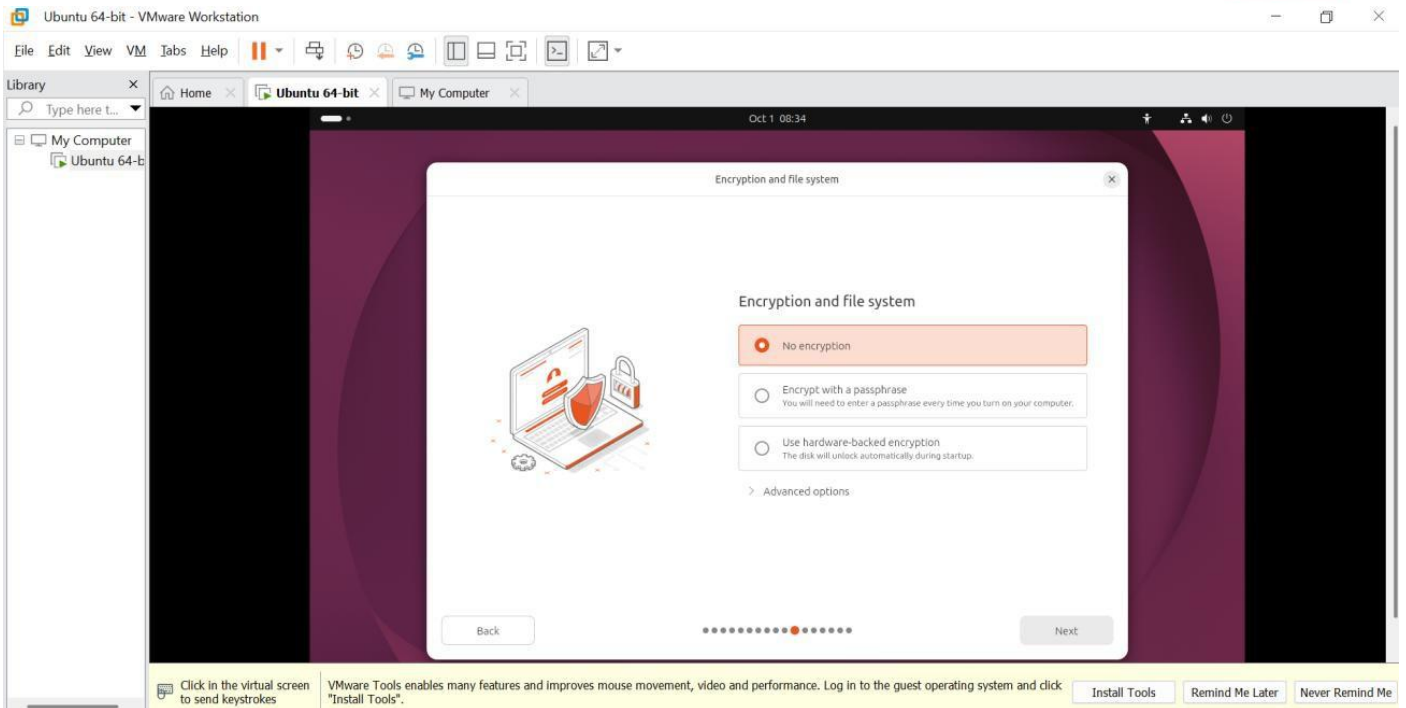


Ubuntu storage configuration screen.

Task 10: Select the Installation Disk

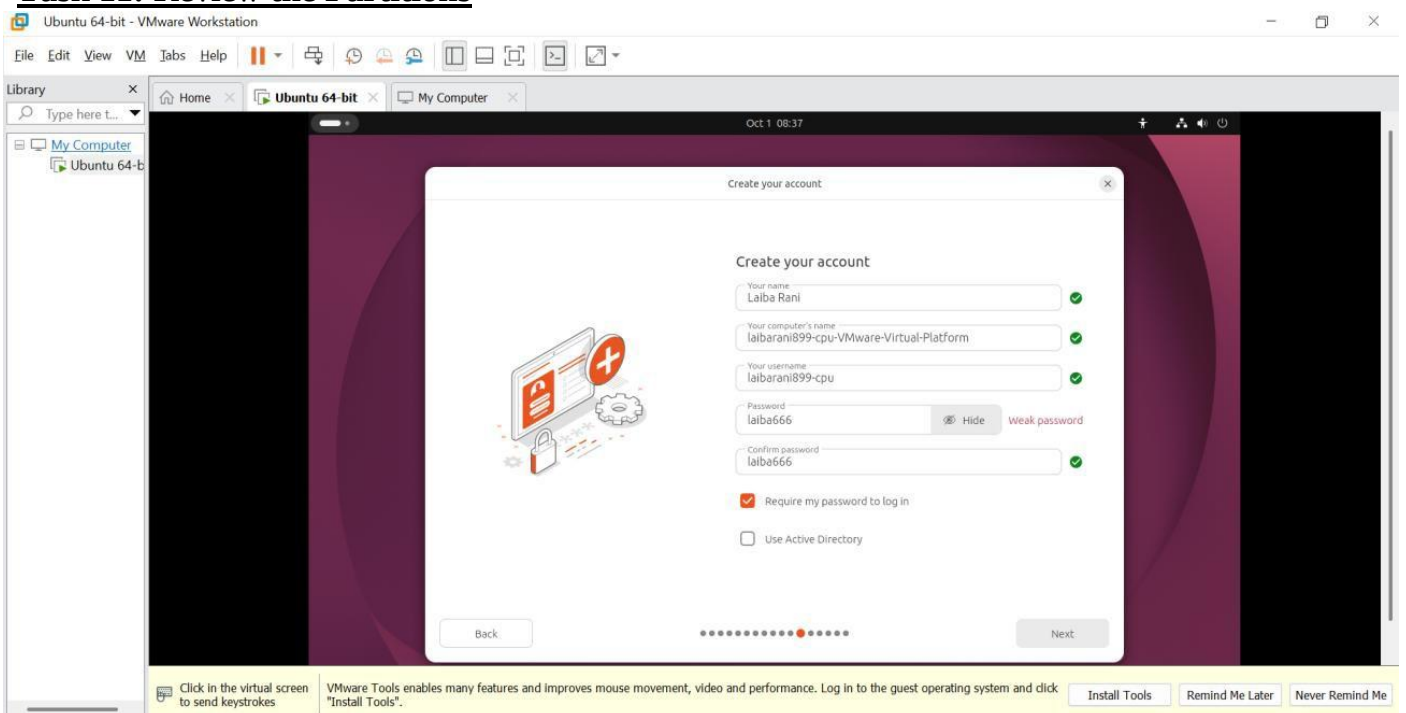


Selected virtual disk for Ubuntu installation.



Virtual disk details shown during installation.

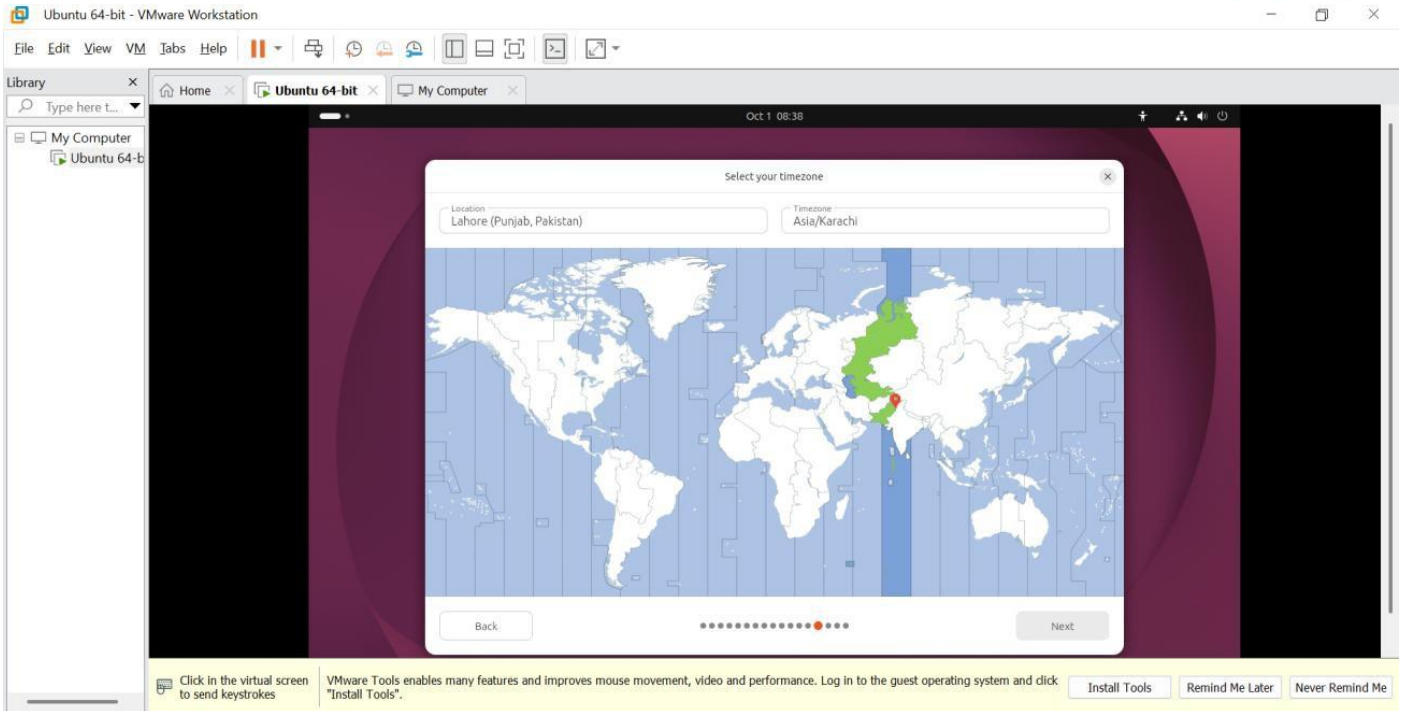
Task 11: Review the Partitions



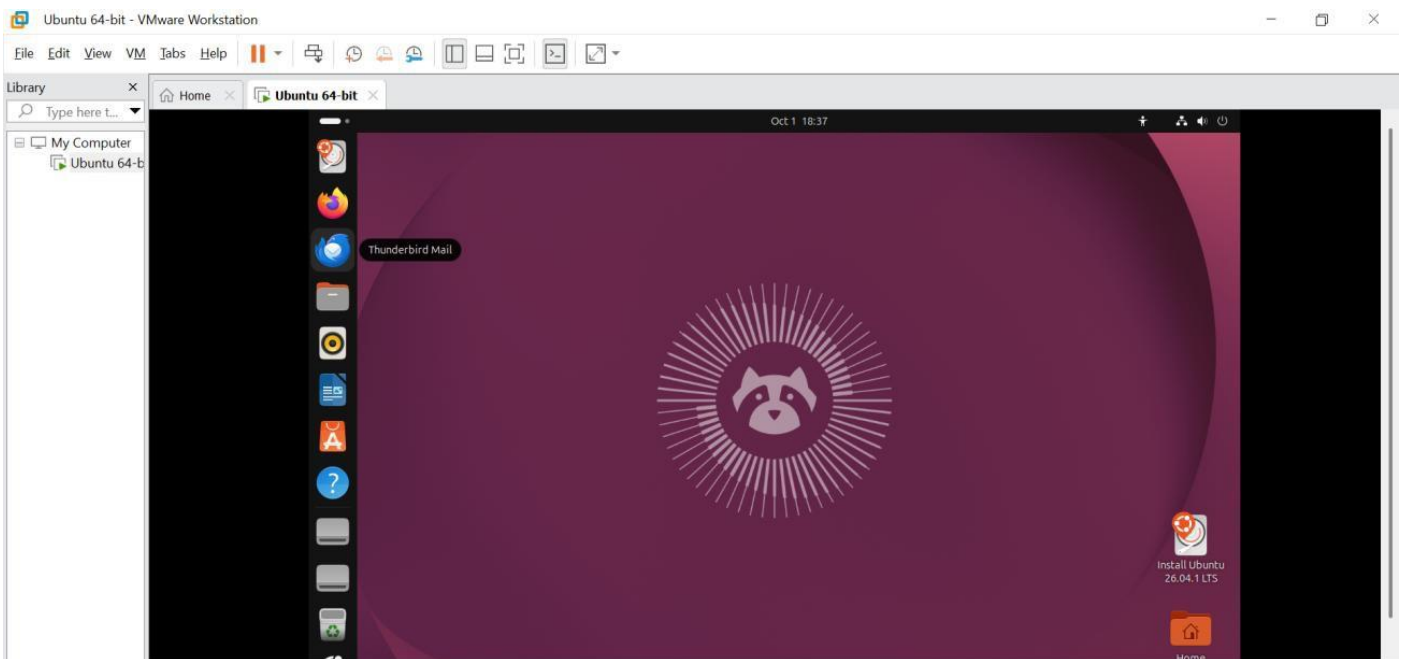
Ubuntu partition layout before installation.



Task 12: Confirm the Storage Changes



Ubuntu storage warning before applying disk changes.



Confirmation of the Ubuntu storage configuration.

**Task 13: Configure the Ubuntu Profile**

New Virtual Machine Wizard ✕

Easy Install Information
This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

User name:

Password:

Confirm:

Ubuntu setup screen showing the configured actual name.



New Virtual Machine Wizard ✕

Easy Install Information
This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

User name:

Password:

Confirm:

Caption: Ubuntu setup screen showing the configured server name.



New Virtual Machine Wizard ✕

Easy Install Information
This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

User name:

Password:

Confirm:

Ubuntu setup screen showing the GitHub username as the username.



New Virtual Machine Wizard ✕

Easy Install Information
This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

User name:

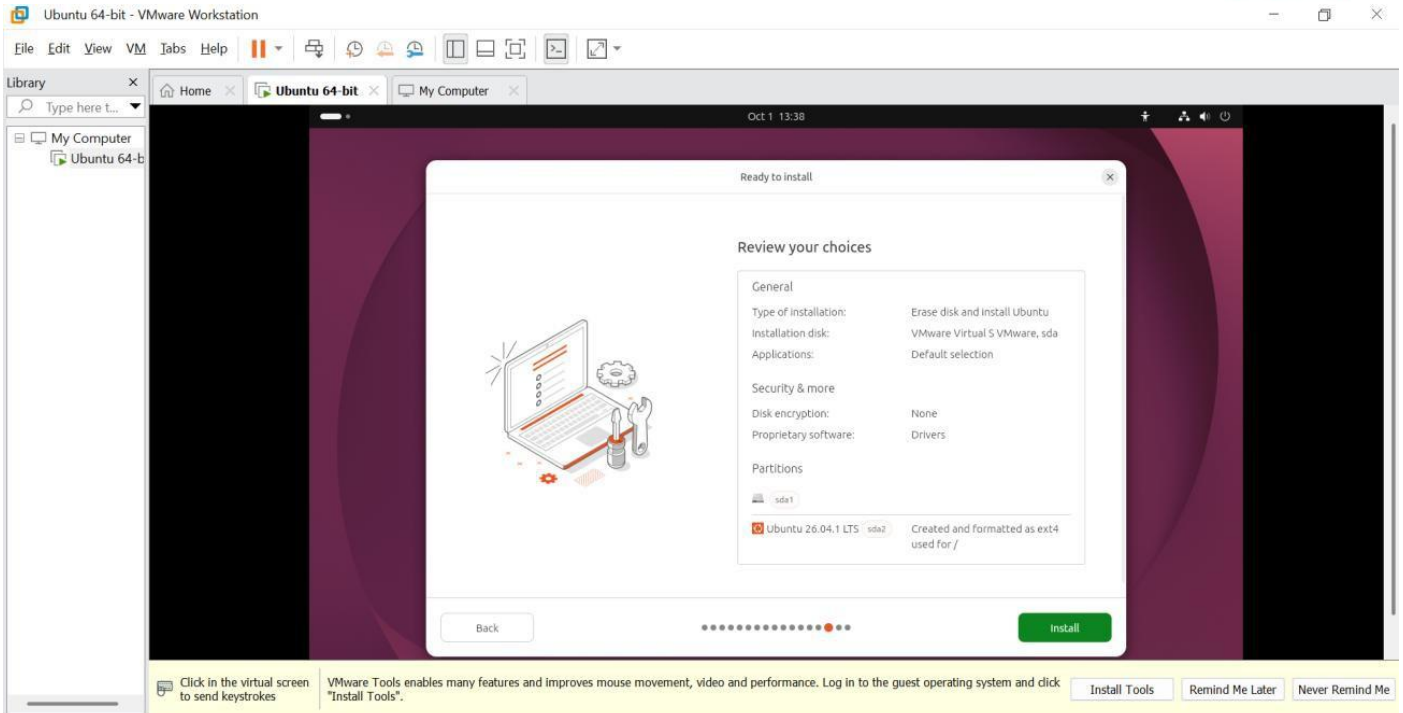
Password:

Confirm:

Ubuntu profile setup information.

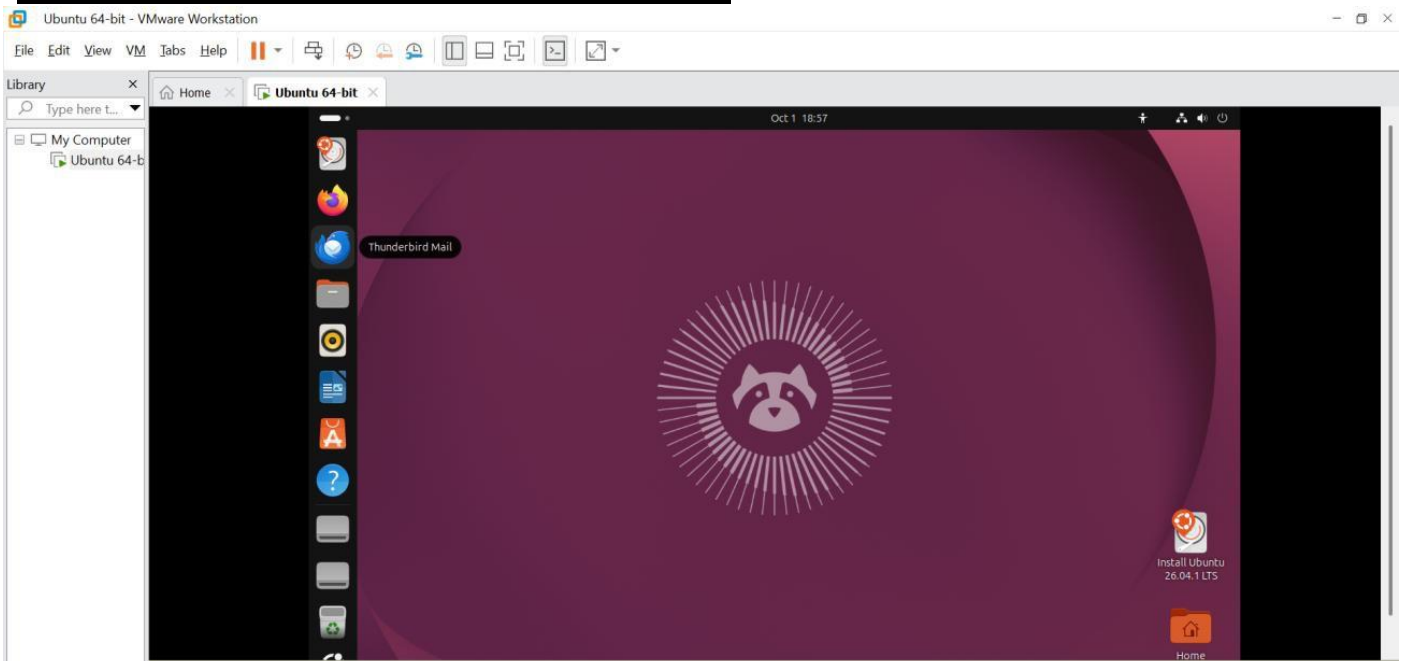


Task 14: Complete the Software Installation

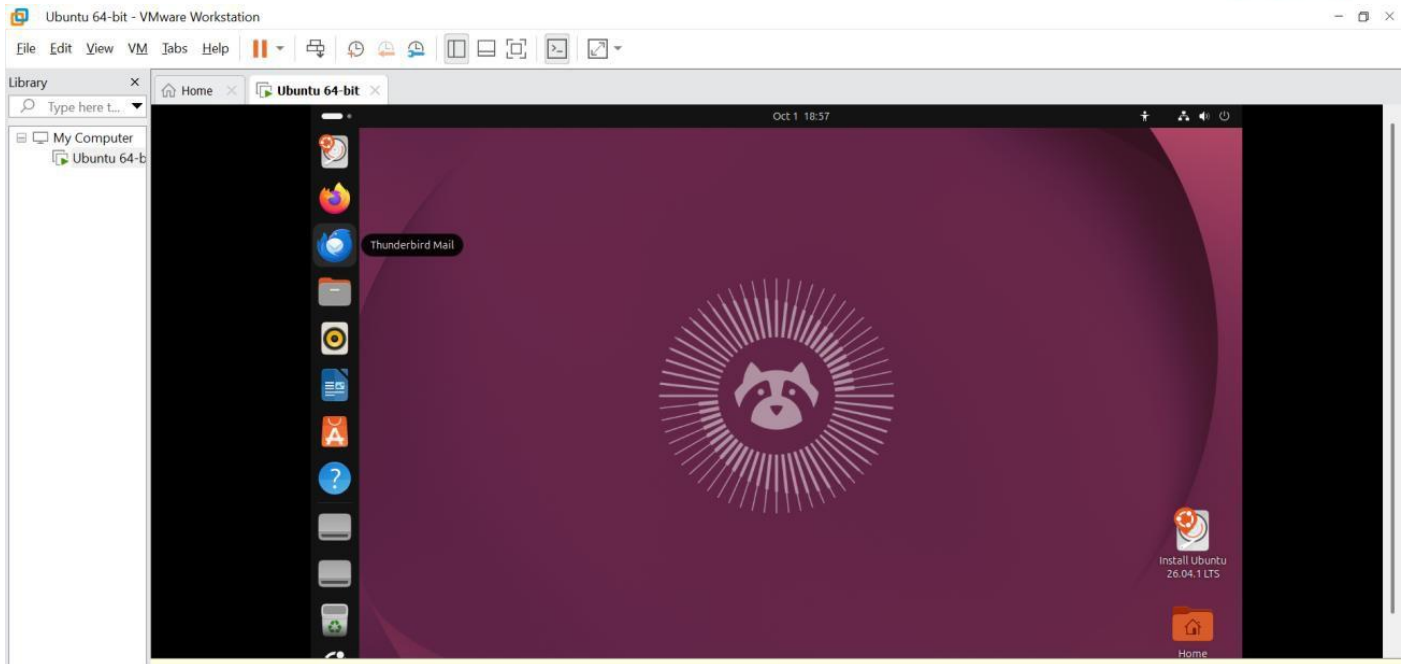


Ubuntu installation progress screen.

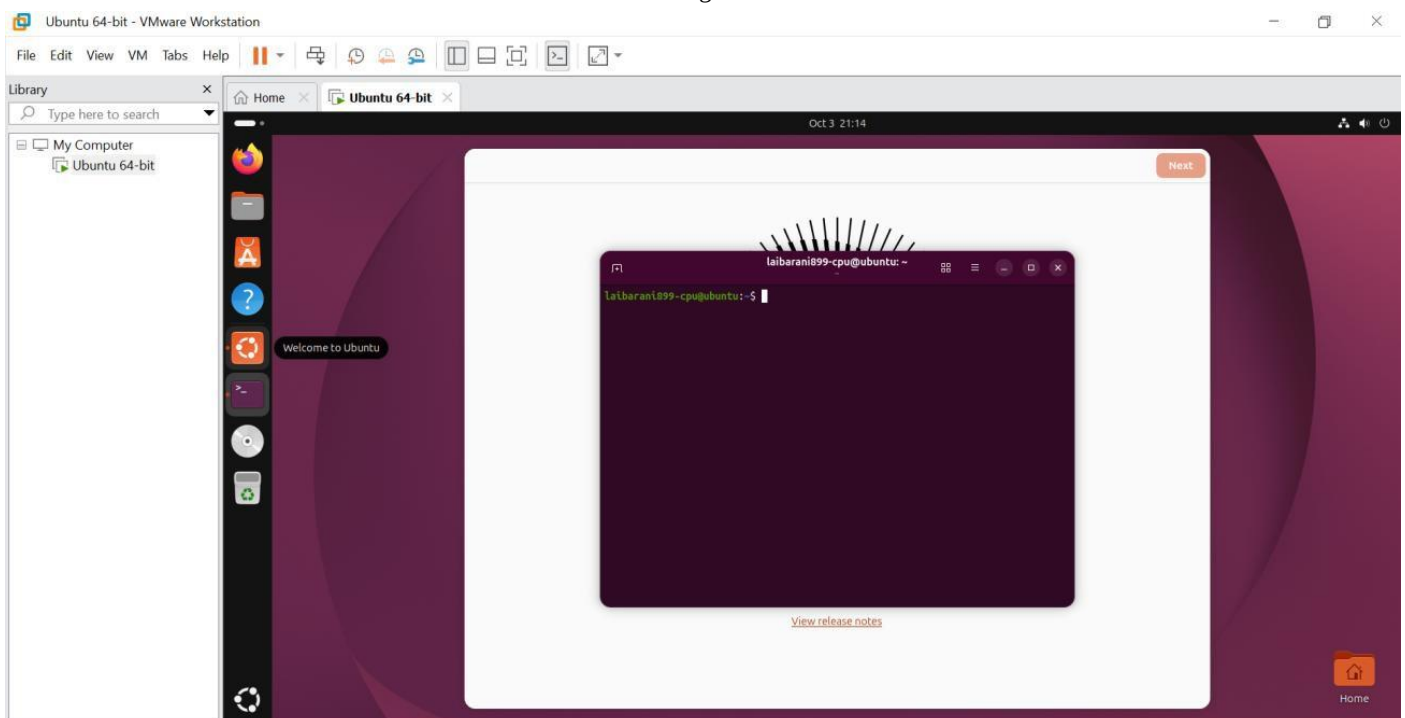
Task 15: Reboot and Verify the Ubuntu Identity



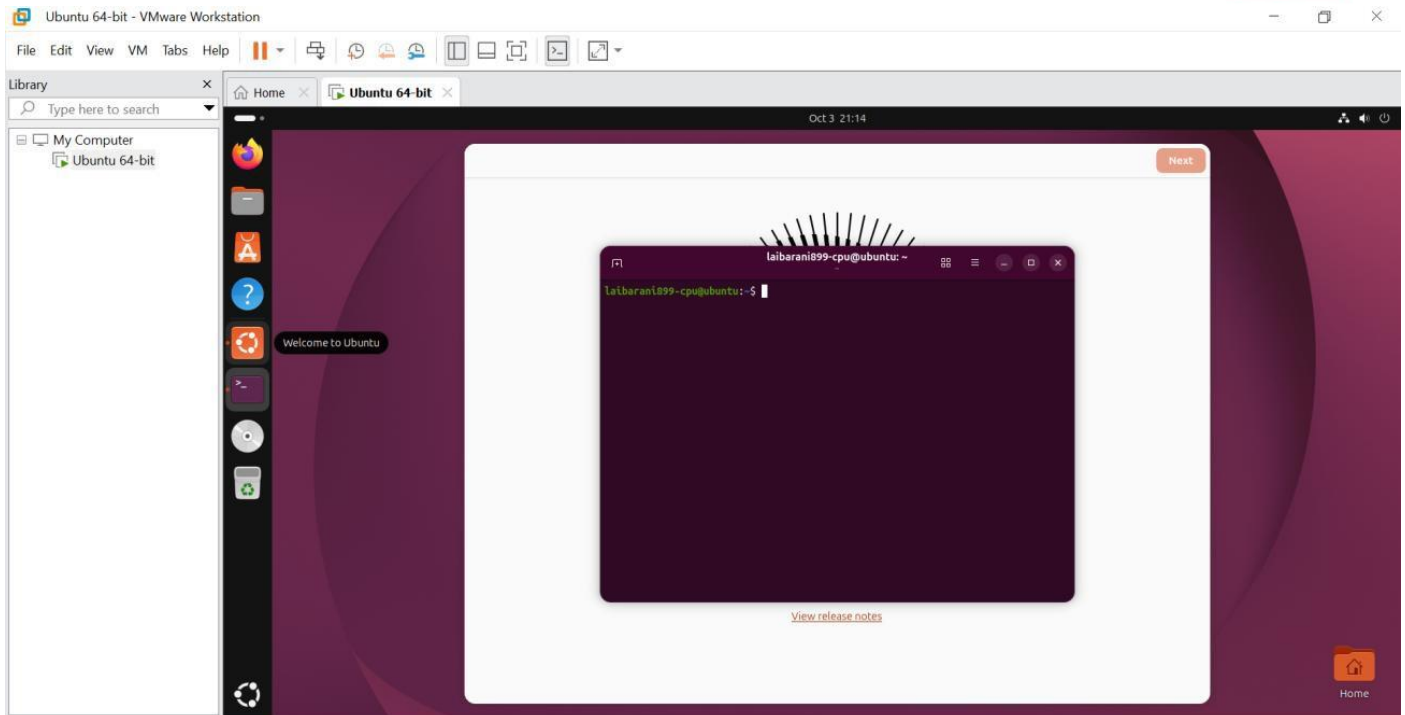
Ubuntu installation completion screen.



Ubuntu login screen.



Ubuntu terminal after login.



Terminal identity verification showing the configured username and hostname.

**Task 16: Find the Ubuntu Server IP Address**

```
laibarani899-cpu@ubuntu: ~  
laibarani899-cpu@ubuntu:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:af:e6:89 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.88.128/24 brd 192.168.88.255 scope global dynamic noprefixroute ens33  
        valid_lft 975sec preferred_lft 975sec  
    inet6 fe80::20c:29ff:feaf:e689/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
laibarani899-cpu@ubuntu:~$
```

Caption: The ip addr command output in Ubuntu.



```

laibarani899-cpu@ubuntu: ~
laibarani899-cpu@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:af:e6:89 brd ff:ff:ff:ff:ff:ff
    inet 192.168.88.128/24 brd 192.168.88.255 scope global dynamic noprefixroute ens33
        valid_lft 975sec preferred_lft 975sec
    inet6 fe80::20c:29ff:feaf:e689/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
laibarani899-cpu@ubuntu:~$

```

Ubuntu IPv4 address identified from the active network interface.

Task 17: Connect to Ubuntu Using SSH

```

PS C:\WINDOWS\System32> ssh laibarani899-cpu@192.168.88.128
The authenticity of host '192.168.88.128 (192.168.88.128)' can't be established.
ED25519 key fingerprint is SHA256:FcCey/MxyxDh6u5GkCNlJu9D5GRQTpEDfWVildfmt0A.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.88.128' (ED25519) to the list of known hosts.
laibarani899-cpu@192.168.88.128's password:

```

Windows PowerShell showing the SSH command.

```

PS C:\WINDOWS\System32> ssh laibarani899-cpu@192.168.88.128
The authenticity of host '192.168.88.128 (192.168.88.128)' can't be established.
ED25519 key fingerprint is SHA256:FcCey/MxyxDh6u5GkCNlJu9D5GRQTpEDfWVildfmt0A.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.88.128' (ED25519) to the list of known hosts.
laibarani899-cpu@192.168.88.128's password:

```

SSH host fingerprint confirmation in Windows PowerShell.



```
laibarani899-cpu@192.168.88.128's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)
```

```
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro
```

```
Expanded Security Maintenance for Applications is not enabled.
```

```
61 updates can be applied immediately.  
5 of these updates are standard security updates.  
To see these additional updates run: apt list --upgradable
```

```
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

```
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.
```

Successful SSH login from Windows into Ubuntu.

```
laibarani899-cpu@ubuntu:~$  
laibarani899-cpu@ubuntu:~$ whoami  
laibarani899-cpu  
laibarani899-cpu@ubuntu:~$ hostname  
ubuntu  
laibarani899-cpu@ubuntu:~$ |
```

Remote Ubuntu identity verified with username and hostname.